

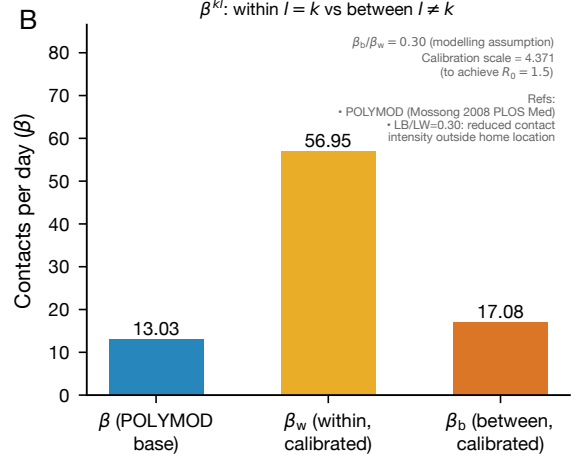
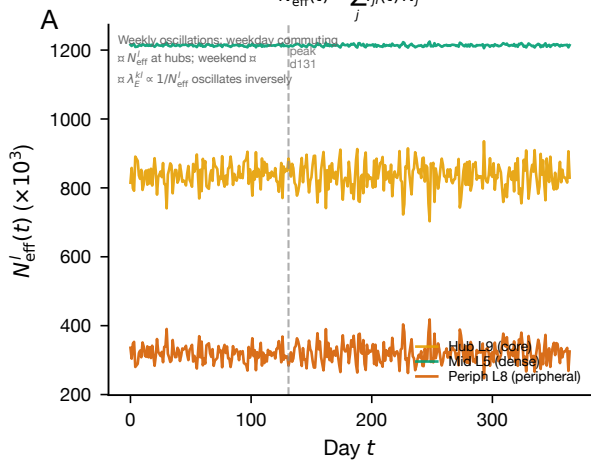
Ingredient 1: effective density

$$N_{\text{eff}}^l(t) = \sum_j f_{jl}(t) N_j$$

Three-ingredient decomposition of $\lambda_E^{kl}(t, a_E) = \beta^{kl} p(a_E) / N_{\text{eff}}^l(t)$ [Eq. 6]

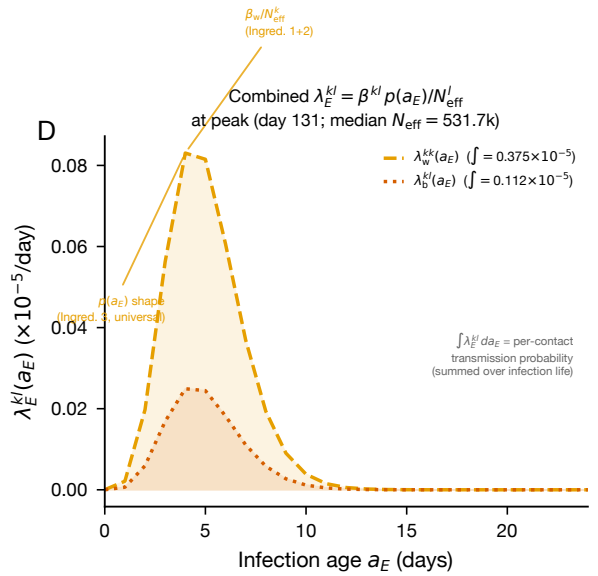
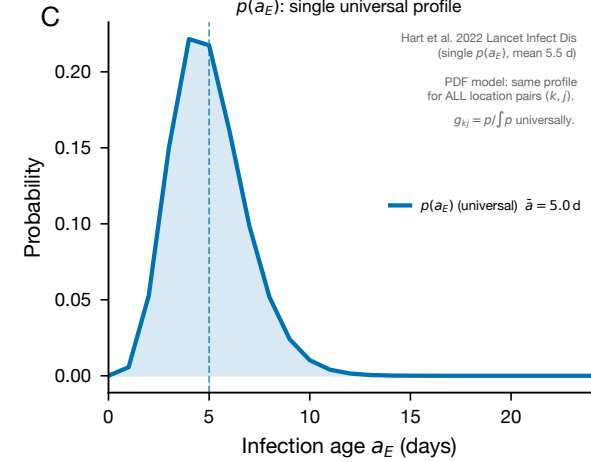
Ingredient 2: location-pair contact rate

β^{kl} : within $l = k$ vs between $l \neq k$



Ingredient 3: biological infectiousness

$p(a_E)$: single universal profile



Kernel $K_{KJ}(t_{pk}, a_E)$ — representative pairs

$$\int K_{KJ} da_E = R_{KJ}(t_{pk}) \text{ (day 131)}$$

